

Exp1:

To implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Data set 1:

Sky	AirTemp	Humidity	Wind	Water	Forecast	PlayTennis
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Same	Yes
Rainy	Warm	Normal	Weak	Warm	Same	No
Sunny	Warm	Normal	Weak	Warm	Same	Yes

Code:

```
import pandas as pd

# Training Dataset

data = [

    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes'],
    ['Sunny', 'Warm', 'Normal', 'Weak', 'Warm', 'Same', 'No'],
    ['Sunny', 'Warm', 'Normal', 'Weak', 'Warm', 'Same', 'Yes']

]

# Column Names
```

```

columns = ['Sky', 'AirTemp', 'Humidity', 'Wind', 'Water', 'Forecast', 'PlayTennis']
# Create DataFrame
df = pd.DataFrame(data, columns=columns)
print("Training Dataset:\n")
print(df)
# Initialize hypothesis with the first positive example
hypothesis = None
for index, row in df.iterrows():
    if row['PlayTennis'] == 'Yes':
        hypothesis = row[:-1].tolist()
        break
# Apply FIND-S Algorithm
for index, row in df.iterrows():
    if row['PlayTennis'] == 'Yes':
        attributes = row[:-1].tolist()

        for i in range(len(hypothesis)):
            if hypothesis[i] != attributes[i]:
                hypothesis[i] = '?'

print("\nFinal Hypothesis:")
print(hypothesis)

```

output:

```
• Training Dataset:

    Sky AirTemp Humidity    Wind Water Forecast PlayTennis
0  Sunny   Warm  Normal  Strong  Warm   Same      Yes
1  Sunny   Warm   High  Strong  Warm   Same      Yes
2  Rainy   Cold   High  Strong  Warm  Change     No
3  Sunny   Warm   High  Strong  Cool  Change     Yes
4  Sunny   Warm  Normal  Weak   Warm   Same     No
5  Sunny   Warm  Normal  Weak   Warm   Same     Yes

Final Hypothesis:
['Sunny', 'Warm', '?', '?', '?', '?']
```

Data set 2:

Income	CreditScore	Employment	Property	Age	LoanApproved
High	Good	Permanent	Yes	Young	Yes
High	Good	Permanent	No	Middle	Yes
Low	Poor	Temporary	No	Young	No
Medium	Good	Permanent	Yes	Middle	Yes
High	Average	Temporary	Yes	Old	No
High	Good	Permanent	Yes	Middle	Yes

Code:

```
import pandas as pd
```

```
# Loan Approval Dataset
```

```
data = [
    ['High', 'High', 'Permanent', 'Good', 'Yes'],
    ['Low', 'Good', 'Temporary', 'Average', 'No'],
```

```

    ['High', 'High', 'Permanent', 'Good', 'Yes'],
    ['Medium', 'Average', 'Permanent', 'Poor', 'No'],
    ['High', 'Good', 'Temporary', 'Good', 'Yes']
]

# Column Names
columns = ['Income', 'CreditScore', 'Employment', 'Property',
'LoanApproved']

# Create DataFrame
df = pd.DataFrame(data, columns=columns)

print("Training Dataset:\n")
print(df)

# Initialize hypothesis with the first positive example
hypothesis = None

for index, row in df.iterrows():
    if row['LoanApproved'] == 'Yes':
        hypothesis = row[:-1].tolist()
        break

# Apply FIND-S Algorithm
for index, row in df.iterrows():
    if row['LoanApproved'] == 'Yes':
        attributes = row[:-1].tolist()

        for i in range(len(hypothesis)):
            if hypothesis[i] != attributes[i]:
                hypothesis[i] = '?'

```

```
# Display Final Hypothesis
```

```
print("\nFinal Hypothesis:")
```

```
print(hypothesis)
```

Output:

```
.. Training Dataset:

   Income CreditScore Employment Property LoanApproved
0   High         High  Permanent     Good           Yes
1   Low          Good  Temporary  Average          No
2   High         High  Permanent     Good           Yes
3 Medium      Average  Permanent     Poor           No
4   High         Good  Temporary     Good           Yes

Final Hypothesis:
['High', '?', '?', 'Good']
```

Data set 3:

CGPA	Communication	Internship	Programming	Aptitude	Placed
High	Good	Yes	Good	High	Yes
High	Excellent	Yes	Good	High	Yes
Medium	Average	No	Average	Medium	No
High	Good	Yes	Excellent	High	Yes
Low	Poor	No	Average	Low	No
High	Good	Yes	Good	Medium	Yes

Code:

```
import pandas as pd
```

```
# Student Placement Dataset
```

```
data = [  
    ['High', 'Good', 'Yes', 'Good', 'High', 'Yes'],  
    ['High', 'Excellent', 'Yes', 'Good', 'High', 'Yes'],  
    ['Medium', 'Average', 'No', 'Average', 'Medium', 'No'],  
    ['High', 'Good', 'Yes', 'Excellent', 'High', 'Yes'],  
    ['Low', 'Poor', 'No', 'Average', 'Low', 'No'],  
    ['High', 'Good', 'Yes', 'Good', 'Medium', 'Yes']  
]
```

```
# Column Names
```

```
columns = [  
    'CGPA',  
    'Communication',  
    'Internship',  
    'Programming',  
    'Aptitude',  
    'Placed'  
]
```

```
# Create DataFrame
```

```
df = pd.DataFrame(data, columns=columns)
```

```
print("Training Dataset:\n")
print(df)

# Initialize hypothesis with the first positive example
hypothesis = None

for index, row in df.iterrows():
    if row['Placed'] == 'Yes':
        hypothesis = row[:-1].tolist()
        break

# FIND-S Algorithm
for index, row in df.iterrows():
    if row['Placed'] == 'Yes':
        attributes = row[:-1].tolist()

        for i in range(len(hypothesis)):
            if hypothesis[i] != attributes[i]:
                hypothesis[i] = '?'

# Display Final Hypothesis
print("\nFinal Hypothesis:")
print(hypothesis)
```

Output:

```
... Training Dataset:
```

	CGPA	Communication	Internship	Programming	Aptitude	Placed
0	High	Good	Yes	Good	High	Yes
1	High	Excellent	Yes	Good	High	Yes
2	Medium	Average	No	Average	Medium	No
3	High	Good	Yes	Excellent	High	Yes
4	Low	Poor	No	Average	Low	No
5	High	Good	Yes	Good	Medium	Yes

```
Final Hypothesis:
```

```
['High', '?', 'Yes', '?', '?']
```

Data set 4:

Fever	Cough	Headache	BodyPain	Fatigue	Disease
-------	-------	----------	----------	---------	---------

Yes	Yes	Yes	Yes	Yes	Positive
Yes	Yes	No	Yes	Yes	Positive
No	Yes	Yes	No	No	Negative
Yes	Yes	Yes	No	Yes	Positive
No	No	Yes	Yes	No	Negative
Yes	Yes	No	No	Yes	Positive

Code:

```
import pandas as pd
```

```
# Disease Diagnosis Dataset
```

```
data = [  
    ['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Positive'],
```

```
['Yes', 'Yes', 'No', 'Yes', 'Yes', 'Positive'],  
['No', 'Yes', 'Yes', 'No', 'No', 'Negative'],  
['Yes', 'Yes', 'Yes', 'No', 'Yes', 'Positive'],  
['No', 'No', 'Yes', 'Yes', 'No', 'Negative'],  
['Yes', 'Yes', 'No', 'No', 'Yes', 'Positive']  
]
```

Column Names

```
columns = [  
    'Fever',  
    'Cough',  
    'Headache',  
    'BodyPain',  
    'Fatigue',  
    'Disease'  
]
```

Create DataFrame

```
df = pd.DataFrame(data, columns=columns)
```

```
print("Training Dataset:\n")
```

```
print(df)
```

Initialize hypothesis with the first positive example

```
hypothesis = None
```

```
for index, row in df.iterrows():  
    if row['Disease'] == 'Positive':  
        hypothesis = row[:-1].tolist()  
        break
```

```
# FIND-S Algorithm
```

```
for index, row in df.iterrows():  
    if row['Disease'] == 'Positive':  
        attributes = row[:-1].tolist()  
  
        for i in range(len(hypothesis)):  
            if hypothesis[i] != attributes[i]:  
                hypothesis[i] = '?'
```

```
# Display Final Hypothesis
```

```
print("\nFinal Hypothesis:")  
print(hypothesis)
```

Output:

```
... Training Dataset:

    Fever Cough Headache BodyPain Fatigue  Disease
0   Yes   Yes   Yes     Yes     Yes   Positive
1   Yes   Yes   No      Yes     Yes   Positive
2   No    Yes   Yes     No      No    Negative
3   Yes   Yes   Yes     No      Yes   Positive
4   No    No    Yes     Yes     No    Negative
5   Yes   Yes   No      No      Yes   Positive

Final Hypothesis:
['Yes', 'Yes', '?', '?', 'Yes']
```

Data set 5:

ContainsLink	ContainsOffer	UnknownSender	Attachment	ManyRecipients	Spam
Yes	Yes	Yes	No	Yes	Yes
Yes	Yes	Yes	Yes	Yes	Yes
No	No	No	No	No	No
Yes	No	Yes	No	Yes	Yes
No	Yes	No	Yes	No	No
Yes	Yes	Yes	No	No	Yes

Code:

```
import pandas as pd
```

```
# Email Spam Detection Dataset
```

```
data = [  
    ['Yes', 'Yes', 'Yes', 'No', 'Yes', 'Yes'],
```

```
['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Yes'],  
['No', 'No', 'No', 'No', 'No', 'No'],  
['Yes', 'No', 'Yes', 'No', 'Yes', 'Yes'],  
['No', 'Yes', 'No', 'No', 'Yes', 'No'],  
['Yes', 'Yes', 'Yes', 'No', 'No', 'Yes']  
]
```

```
# Column Names
```

```
columns = [  
    'ContainsLink',  
    'ContainsOffer',  
    'UnknownSender',  
    'Attachment',  
    'ManyRecipients',  
    'Spam'  
]
```

```
# Create DataFrame
```

```
df = pd.DataFrame(data, columns=columns)
```

```
print("Training Dataset:\n")
```

```
print(df)
```

```
# Initialize hypothesis with the first positive example
```

```
hypothesis = None
```

```
for index, row in df.iterrows():  
    if row['Spam'] == 'Yes':  
        hypothesis = row[:-1].tolist()  
        break
```

```
# FIND-S Algorithm
```

```
for index, row in df.iterrows():  
    if row['Spam'] == 'Yes':  
        attributes = row[:-1].tolist()  
  
        for i in range(len(hypothesis)):  
            if hypothesis[i] != attributes[i]:  
                hypothesis[i] = '?'
```

```
# Display Final Hypothesis
```

```
print("\nFinal Hypothesis:")  
print(hypothesis)
```

Output:

... Training Dataset:

	ContainsLink	ContainsOffer	UnknownSender	Attachment	ManyRecipients	Spam
0	Yes	Yes	Yes	No	Yes	Yes
1	Yes	Yes	Yes	Yes	Yes	Yes
2	No	No	No	No	No	No
3	Yes	No	Yes	No	Yes	Yes
4	No	Yes	No	No	Yes	No
5	Yes	Yes	Yes	No	No	Yes

Final Hypothesis:

['Yes', '?', 'Yes', '?', '?']